

UROP End of Semester Essay (Spring 2025) – Ottavia Personeni

Last semester, I started the SPL Electrospray Thruster Fabrication project by fabricating and characterizing four complete thruster frames using Varapor 100, a porous material with 100 nm pores ideal for the precise tip geometries we needed. I learned a glass-cutting technique and found that alternating deionized-water soaking with light drying minimized cracking. I developed a new sanding and polishing routine that produced more uniform etching and fewer deformations. Throughout the semester, I documented each procedure, success, and setback to refine the workflow and improve overall thruster performance.

In parallel, I learned to operate the VHX microscope to capture high-resolution images of emitters before firing, and I set up thruster frames in temporary mounts for TVAC runs. I practiced the thruster startup procedure of gradually ramping voltages and running square and triangle wave firing tests on the thruster. The tests provided good data, but the TVAC TOF scan was not successful due to the way the thruster was secured in the temporary mount. However, these experiences still prepared me to transition from fabrication and practice runs into the upcoming phase of firing and data.

At the end of last semester, my plan was to shift my focus to performing vacuum tests and characterizing ion beams for lithium salt ionic liquids. These liquids have been hypothesized, and potentially proven, to yield almost 100% lithium-ion streams. My goal was to establish a reproducible, well-documented fabrication-to-test pipeline, so that I could analyze performance data based on geometries and propellants.

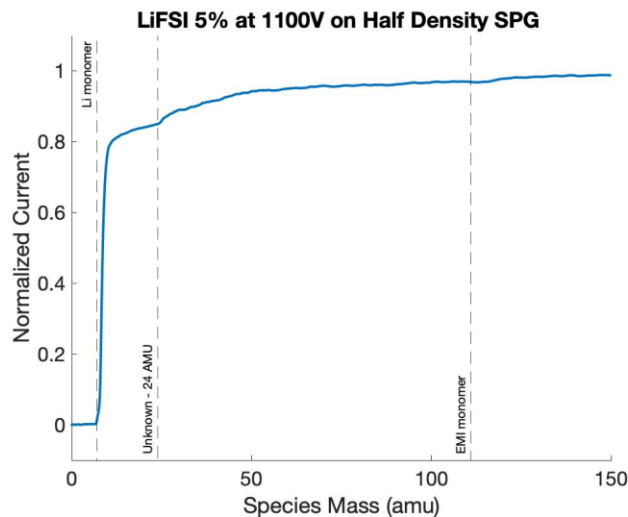


Figure 1: Original SPL vacuum test that showed a 100% lithium-ion stream.

I revisited our original thruster with plans for increased emitter wetting. By increasing the ionic liquid droplets added to the filter papers within the temporary mount, the amount of liquid coverage on the emitter increased. This allowed for an easier and quicker startup procedure, as

well as higher current readings and better overall performance. I brought these measurements into MATLAB for a first pass look to confirm that the data was being taken correctly and that the readings were expected.

Next, I set out to build a new thruster temporary mount to test out the lithium salt ionic liquid (LiFSI 5%) solution that had been previously created and stored in the lab. Using the frame from last semester's tests, I began to assemble and prepare for a new run. I began by cleaning the original thruster using the SPL documented cleaning routine.

The frame was already on the thruster, since it was used for previous tests. The last step of the process included using the plasma etch machine, which provided a critical error. In cleaning the thruster, I did not wait until the glass was *completely* dry before putting it into the machine. Therefore, the water caused the thruster to get much hotter than expected, leading to intense cracking.

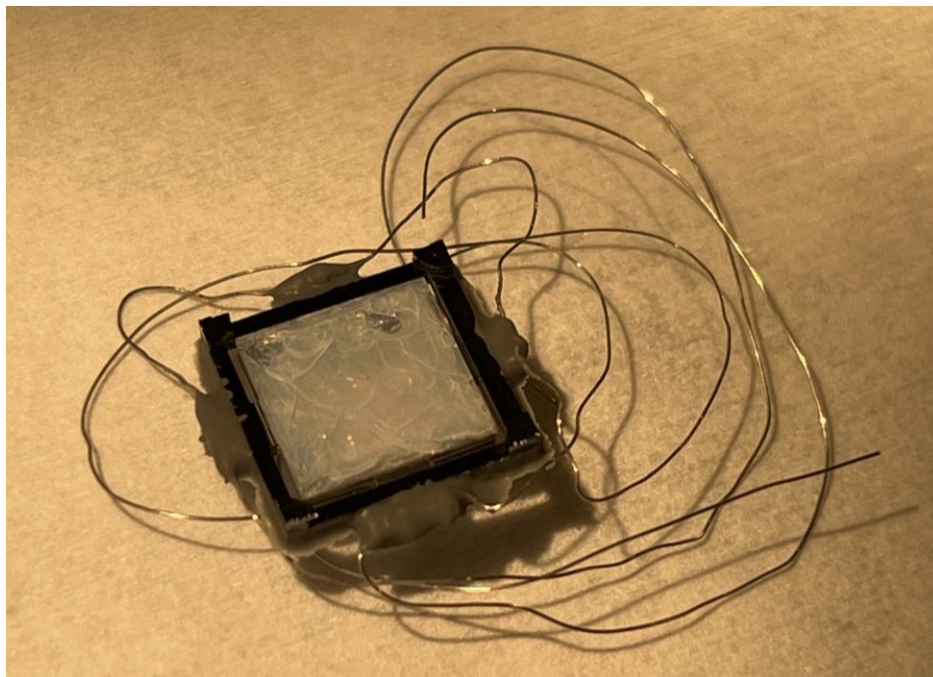


Figure 2: Cracking after the plasma etch machine due to incomplete drying of the glass.

Thankfully, the thruster was salvageable because the crack did not directly interfere with the single emitter in the center of the glass. Through Matt's guidance, I learned to use the SEM (scanning electron microscope), which showed us (in high resolution) that the single emitter was in fact unharmed. However, the glass did become extremely fragile, which posed even more challenges.

At this point, the extractor needed to be attached, yet there were no extractor grids currently in the lab that perfectly aligned with the emitter. Eventually, we were able to center the grid over

the emitter under the microscope. On the electrical side, I attempted to repair the fragile and sometimes broken heater connection. The soldering was difficult and often did not hold, so Matt took over this task.

At this point, the temporary mount setup began, yet, in the routine stacking of the filter papers under the thruster, the glass broke completely. Even with extreme care and very little force, the glass was fragile enough to shatter in a second, destroying a thruster I'd spent weeks preparing. At this point, it was not salvageable. These setbacks reinforced the need for strict drying protocols and ultra-careful handling before high-temperature or vacuum processes.

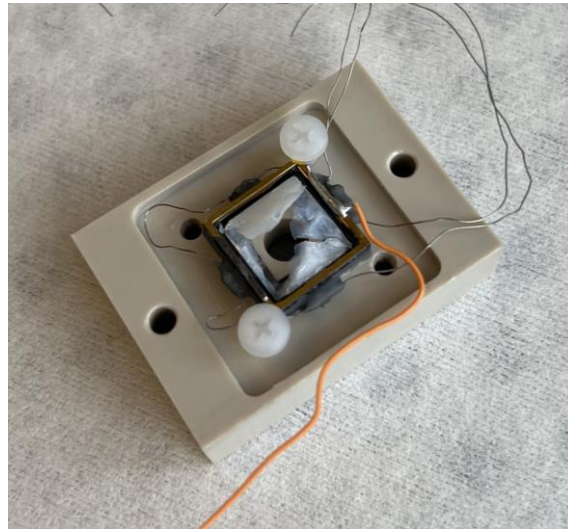


Figure 3: Shattering during the filter paper stacking of the temporary mount setup process.

To finish off the semester, I began working on making Varapor 100 glass squares for the thrusters using the scraps of previous glass cutting work. These squares are about 4 x 4 mm, sanded down to a 1 mm thickness, and polished evenly. In lab, there are now thirteen 4 x 4 mm (± 0.1 mm) squares and five 4 x 4 mm (± 0.5 mm). These many uniform smaller pieces of glass can be paired with an adaptor that will center the glass into the thruster frame. These smaller glass squares are big enough for a single emitter and are a great way to take advantage of scraps.



Figure 4: Varapor 100 squares and scraps — in petri dishes stored in building 35's lab.